

Education

Ontario Tech University, Oshawa, ON

Bachelor of Engineering – Software Engineering | Expected June 2027

- Relevant coursework: Software Design, Data Structures & Algorithms, Operating Systems, Control Systems, and Data management.
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Work Experience

MyHomeworkRewards Hybrid

Course Developer & Web Programmer | Feb 2025 – April 2025

- Developed interactive, curriculum-aligned Grade 12 Data Management lessons, **increasing student engagement by 30%**.
- Built responsive, accessible lesson pages **using HTML, CSS, Bootstrap, and JavaScript, improving usability by 25%**.
- Designed visual elements and embedded questions that boosted student interaction **rates by 35%**.
- Prioritized accessibility and intuitive layout, contributing to an edtech platform that boosted overall **learning satisfaction scores by 20% among Ontario students**.
- **Implemented version control and project management** using Git/GitHub to streamline development and updates.

Research Student (Part-Time, Hybrid)

Smart Energy Systems, Faculty of Engineering and Applied Science | December 2025 – March 2026

- Designed and developed a resilience assessment calculator for hybrid interconnected infrastructure systems using probabilistic and simulation-based modeling techniques.
- Implemented modular models integrating wind-field decay analysis, lognormal fragility curves, Monte Carlo simulation, recovery modeling, and resilience metrics.
- Conducted literature reviews on infrastructure interdependencies, extreme-event modeling, resilience frameworks, and sustainable energy systems to support research validation and system development.
- Performed case study analyses to evaluate cascading failures, outage propagation behavior, recovery performance, and critical infrastructure vulnerabilities.

- Developed simulation logic for Inherent Resiliency Layers (IRL), adaptive recovery sequencing, and interconnected subsystem failure analysis.
 - Created technical diagrams, activity diagrams, sequence diagrams, and computational workflow documentation for academic presentations and research communication.
 - Assisted with MATLAB-based modeling, resilience simulations, and technical research documentation.
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Research Student (Full-Time, Hybrid)

Smart Energy Systems, Faculty of Engineering and Applied Science | May 11, 2026 – present

- Running and evaluating **RE-LLM** and **MG-OPT** systems to support research in resilient and integrated smart energy systems.
- Defining and analyzing case studies for multi-generation energy systems involving hydrogen production, fuel cells, electrolyzers, heating, cooling, clean water, hot water, waste treatment, EVs, and FCVs.
- Conducting extensive literature reviews on integrated energy systems to identify configurations, technologies, system parameters, sizing requirements, and optimization approaches.
- Investigating real-world applications of smart energy systems for residential buildings, farms, industries, ships, seaports, airports, rail systems, and waterfront infrastructure.
- Preparing and supporting simulation workflows using **HOMER** and related optimization tools for performance evaluation and energy system feasibility studies.
- Supporting NRHESS-style resilience and sustainability research through system modeling, scenario analysis, technical documentation, and research presentations.
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Rasam Hybrid

Front-end and back-end developer | December 2025- Feb 2026

- **Designed and implemented full-stack features**, including CRUD operations (Create, Read, Update, Delete)
 - Developed user-friendly front-end components and connected them to **back-end services**
 - Managed application **data flow between client and server**
 - Improved app stability by **troubleshooting bugs and optimizing existing code**
 - Participated in **feature planning and agile development cycles**
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Projects/Competition

Technation Canada's AI Equity Challenge | *Javascript, Gemini, Python, and api backend.*

- Spearheaded the development and pitch of Equity Decoder, a bilingual, policy-aligned Chrome extension that rewrites biased job postings, **earning over 90% positive feedback from students, faculty, and employers during live testing.**
- Developed responsive lesson-style dashboards and real-time analytics visualizations **using HTML, CSS, Bootstrap, and JavaScript**, contributing to a fully functional prototype that detected bias across multiple equity-seeking groups.
- Delivered a 10-act, 5-minute technical pitch that increased audience comprehension and engagement **by 40%**, leading to invitations for mentorship, funding opportunities, and early-stage pilot testing across major Canadian cities.

Monte Carlo Evacuation Simulator | *JavaScript, HTML/CSS, Data Visualization, Simulation Modeling*

- Built a **Monte Carlo–based evacuation simulation system** modeling real-world human behavior, incorporating **stochastic reaction times, queuing dynamics**, and path optimization to evaluate building safety under emergency scenarios
- Developed an **interactive web application** with configurable parameters and **real-time visualization of evacuation metrics**, including worst-case (T_{95}) risk analysis and exceedance probability
- Designed **sensitivity analysis tools** to identify high-impact variables (e.g., occupancy and reaction time variability), **enabling data-driven safety recommendations.**

Distance-Sensing Alert System—Accessibility-Focused Tech Project | *Arduino, Ultrasonic Sensor, LEDs, Buzzer, C/C++*

- Designed a real-time alert system using ultrasonic sensors, LEDs, and a buzzer to assist individuals with visual, hearing, or cognitive impairments.
- Implemented intuitive signals (color-coded LEDs + Morse-like audio patterns) to indicate safe, cautious, or dangerous distances.
- Collaborated on hardware and software integration, demonstrating inclusive design principles and practical problem-solving in accessible technology.

Engineering Demo Fest 2025 – Capacitor-Powered Vehicle, 2nd Place | *Supercapacitor, TT Motor, Diode, Wiring, CAD, Circuit Design*

- Led the electrical design of a supercapacitor-powered vehicle, integrating TT motors, diodes, and circuit wiring for maximum energy efficiency.
- Collaborated with a multidisciplinary team to redesign and rebuild the vehicle under tight deadlines, turning a near-disaster into a 2nd-place finish.
- Applied classroom concepts to a real-world challenge, gaining hands-on experience with circuit design, CAD, and mechanical integration.
- Developed teamwork, problem-solving, and adaptability skills by balancing technical tasks, tight timelines, and cross-functional collaboration in a high-stakes competition.

Hack the Valley—University of Toronto | *React Native, Firebase, and Node.js*

- Built NewLeaf, an AI-powered app helping newcomers navigate life with checklists, tips, and real-time chat.
- Developed front-end and full-stack features using React Native, Node.js, Firebase, and Google Gemini AI.
- Collaborated under pressure at Hack the Valley, delivering an innovative, user-friendly prototype.

Password Manager – Secure Credential Storage | *Java, SHA-256, Git*

- Built a Java password manager with strong validation and secure storage using SHA-256 hashing.
 - Developed user registration and authentication, ensuring safe and reliable credential management.
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Leadership & Volunteering

Peer leader, Orientation leader, Volunteer and executive – Enactus, Cpp north, Women in Eng, CI Tech 2024-present

- Orientation Leader & Peer Mentor (2024–2026): Guided and supported new students through workshops, university adjustment, and academic organization.
 - ENGSoc Director & Women’s Student Association Social Executive: Promoted student engagement, inclusivity, and leadership in STEM initiatives.
 - Discover Engineering 2024 & CPP North Volunteer: Led workshops; mentored students; and assisted with event coordination, building teamwork and communication skills.
 - Hackathon Competitor: Participated in three hackathons, gaining hands-on experience in teamwork, problem-solving, and rapid project development.
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Technical Skills

- **Programming Languages:** Python, Java, C++, C#, javascript,
 - **Frameworks & Tools:** React, Node.js, GitHub
 - **Concepts:** Monte Carlo Simulation, Stochastic Modeling, Data Visualization, Sensitivity Analysis, Queuing Systems, Risk Modeling, Object-Oriented Programming, API Development, Agile SDLC
 - **Cloud & DevOps:** CI/CD Pipelines, Linux/Unix
 - **Databases:** MySQL, MongoDB
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